

Public-Data Analytics on the InGen Robotics Portfolio

Market · Competition · BI · Forecasting · Operations

Capstone report	Data Analyst Internship · Futureonauts Program
Author	Ziheng Wang, Data Analyst Intern
Period	13 weeks · Weeks 1–12 delivered, synthesised here in Week 13
Scope	Five product verticals — Fari (eldercare), Senpai (education), Sentinel Prime AI (indoor security), Aido Rover (outdoor patrol), Aido Humanoid (humanoid)
Basis	Public data only — government statistics, SEC filings, patents, news, search trends, licensed public benchmarks. No InGen internal data was used or requested for any figure in this report.

How to read this report

The **Executive Summary** on the next page stands alone — a senior stakeholder can read it in three minutes and stop there. Sections 2–9 give one workstream each, in the order they were built, and every one cites the notebook or workbook in the repository that produced it. Section 10 is an honest account of what public data could not answer. Section 11 says what to do next.

1. Executive Summary

Over thirteen weeks I built a public-data intelligence picture of InGen's five-vertical robotics portfolio: how large each market is, which are gaining momentum, how competitors actually perform financially, what demand looks like 24 months out, and how two of InGen's own capabilities — anomaly detection and fleet support — behave under analysis. Everything rests on public sources; nothing here required internal access. That is both the method's strength (it is reproducible and shareable) and its limit (Section 10).

Six findings that matter

1. Funding is not a moat.	Across 15 priority peers, capital raised and defensible IP diverge sharply. Figure AI has raised ~\$1.68bn; several better-defended competitors have raised a fraction of that. Week 6's financials confirmed the pattern from the opposite direction: the most profitable public peers (Teradyne 58.5% gross margin, Cognex 68%) are precision-instrument businesses, while the consumer-robotics pure-play (iRobot, 20.9% gross margin, -15.1% operating) is losing money. Margin structure, not headline funding, predicts durability.
2. The biggest near-term market is not the loudest one.	Indoor security (Sentinel Prime AI) has the clearest near-term economics — a guard-replacement value proposition against a costed alternative. Humanoid attracts the capital and the headlines, but published estimates of its market span an order of magnitude (~\$290M to ~\$4.89bn depending on scope and year), so it is a scenario, not a forecast.
3. Attention and opportunity are not aligned.	The demand-signal index ranks Aido Rover first (64) and Sentinel Prime AI last (41) — the exact inverse of the market-attractiveness ranking. Search and news measure <i>attention</i> ; sizing measures <i>money</i> . Sentinel looks like an under-marketed strong asset.
4. Demand momentum is up across all five verticals, and models beat naive baselines.	Best-model MASE is below 1.0 in four of five verticals (education 0.54, humanoid 0.59, outdoor patrol 0.69, indoor security 0.96), i.e. better than a seasonal-naive forecast. No single model wins everywhere — ETS, Prophet and XGBoost each take a vertical.
5. Sentinel's alert problem is solved by persistence, not by a better threshold.	On real public sensor benchmarks, no threshold policy reaches a 5% false-alarm budget at 80% recall — that ceiling is set by model quality. But requiring an anomaly to persist 120 minutes cuts operator alert load 33× (16.35 → 0.50 false alerts/day) while recall <i>rises</i> to 88% and all 4/4 real failures are still caught.
6. Support cycle time is a parts problem, not a people problem.	Parts & dispatch is 53% of all support process time and 100% of it is unstaffed waiting. A regional parts buffer cuts mean cycle time 23% ; adding a field engineer cuts it 8%; rerouting triage — the obvious lever — does nothing measurable.

What I recommend

Commercially: treat Sentinel Prime AI as the near-term revenue engine and market it harder — it is the strongest asset with the weakest attention. Treat humanoid as a real option with scenario-bounded planning, not a forecast. **Operationally:** fund a regional parts buffer before hiring field engineers, and ship Sentinel's anomaly detection with a two-tier alert rule (fast tier for intrusion, sustained tier for degradation). **Analytically:** the highest-leverage unlock is a small slice of real internal data — every caveat in Section 10 dissolves the moment there is something real to calibrate against.

Source in repo: reports/week13/ (this report) · full detail in Sections 2–11 below.

2. Methodology

The work ran in four phases, each building on the last. Phase 1 established what InGen sells and who competes with it, and built the data plumbing. Phase 2 sized the markets and read demand and competitor finances. Phase 3 turned that into a queryable warehouse and dashboards. Phase 4 applied forecasting, anomaly detection and process analytics to InGen use cases.

The 13 weeks

Wk	Workstream	Principal artifact
1	Product & competitor landscape	Product profiles; 40+ competitor landscape; data dictionary
2	Competitive intelligence	15 priority peers with real patents & headcounts
3	Public-data pipeline & warehouse	12-dataset ingestion pipeline; DuckDB warehouse; 15 tests
4	Market sizing	TAM/SAM/SOM, dual-method, tornado sensitivity
5	Demand signals	Search + news + review mining → demand-signal index
6	Financial benchmarking	Public FY2024 financials + private funding, sourced per row
7	SQL & analytics warehouse	Star schema, 100k-row synthetic dataset, 15 queries
8	Dashboards (Tableau, Looker)	Specs, extracts, prototypes, design-review log
9	Power BI & mid-internship review	Exec scorecard, 10 DAX measures, review packet
10	Demand & adoption forecasting	5 models × 5 verticals; Bass diffusion for humanoid
11	Anomaly detection (Sentinel)	5 detectors on 2 licensed public benchmarks
12	Process optimisation	DES of support lifecycle; capacity scenarios
13	Capstone & handoff	This report; exec deck; dashboard pack; HANDOFF.md

Principles held throughout

Sourced or not stated. Every quantitative claim carries a source and a date. Where published estimates disagree — and in this industry they disagree a lot — the report gives a range rather than picking the convenient number.

Synthetic data is always labelled. Weeks 7–9 and 12 use a synthetic warehouse because no internal data was available. It is never presented as real, and every table that rests on it says so.

Reproducible by construction. Fixed seeds, committed manifests, and pytest suites in Weeks 3, 5, 6, 7, 11 and 12. Anyone can re-run the repository and obtain these numbers.

Methods validated where ground truth exists. Week 12's driver regression was checked against the known generating parameters of its own simulation and recovered 7 of 9 effect directions; Week 10's forecasts were scored against naive baselines on held-out data; Week 11's detectors were compared against a control-chart benchmark on labelled public data.

Source in repo: [plan/week01.md](#) ... [plan/week13.md](#) · [README.md](#) · [INDEX.md](#)

3. Market Sizing

Each of the five verticals was sized twice — top-down from published market anchors, and bottom-up from unit economics (addressable units × serviceable share × penetration × average selling price) — and the two were reconciled. Where they disagreed, the gap itself was informative: it usually meant the published figure bundled a broader scope than InGen actually sells into.

What the numbers can and cannot say

Published estimates for these markets vary by an order of magnitude. Humanoid robotics is quoted anywhere from roughly \$290M to \$4.89bn depending on the year and what counts as a humanoid; security robotics ranges from about \$4.7bn to \$19bn depending on whether fixed surveillance is included. **Any single-point market number in this space is a choice of source, not a fact.** The workbook therefore records a range per vertical, names the anchor behind each bound, and keeps the assumption register separate so a reader can substitute their own beliefs and re-derive.

How each number was built

Top-down starts from a published market anchor and carves InGen's addressable slice out of it by geography, segment and product fit. It is fast and it inherits the anchor's assumptions wholesale — including whatever the analyst who wrote it decided a 'security robot' was.

Bottom-up starts from countable things: how many facilities, households or campuses exist (US Census, BLS, NCES, OECD, World Bank), what share is realistically serviceable, what penetration is plausible over the horizon, and what InGen could charge. Every one of those four terms is a named assumption in the register, with a source or an explicit judgement flag.

Reconciliation is where the value is. When bottom-up came in far below a published top-down figure, the difference was almost always scope: the published number counted adjacent categories InGen does not sell into. That gap is a finding, not an error — it means the headline market number overstates the reachable one.

Ranking by near-term attractiveness

Vertical / product	Near-term attractiveness	Why
Sentinel Prime AI — indoor security	Highest	Clear guard-replacement economics against a costed alternative; buyer already has a budget
Fari — eldercare	Medium-high	Large addressable population, but adoption-rate constrained; reimbursement is the gate
Senpai — education	Medium	Steady, but tied to institutional budget cycles
Aido Rover — outdoor patrol	Scenario range	High potential, high uncertainty in serviceable share
Aido Humanoid — humanoid	Scenario range	Early; the market is a bet on future deployment, not a current addressable pool

Sensitivity analysis (one-at-a-time tornado, per vertical) showed the same structure everywhere: **penetration rate dominates**, ASP second, and the addressable-unit count a distant third. That matters strategically — it means the market size is mostly a function of how fast buyers adopt, which is a go-to-market variable InGen partly controls, rather than a fixed external quantity.

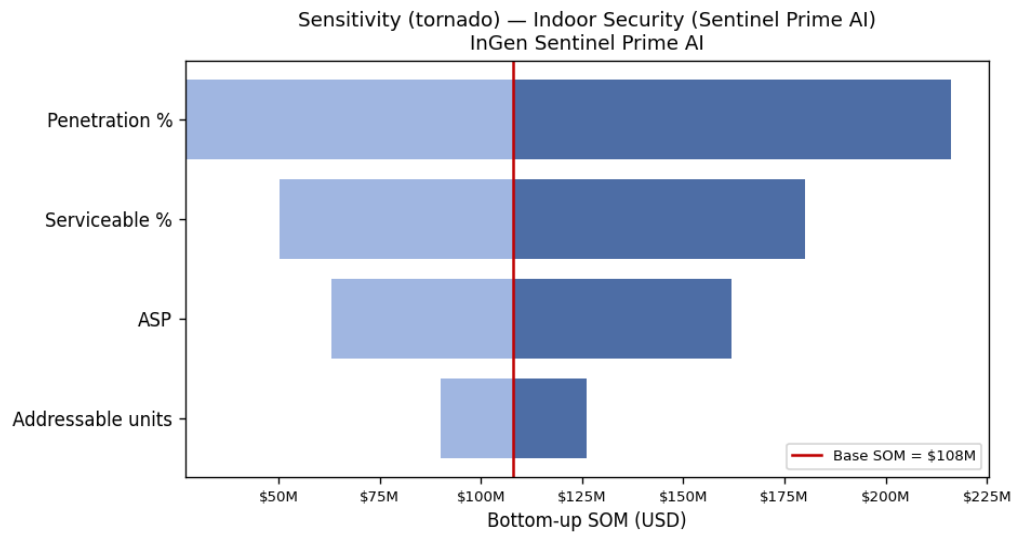


Figure 3.1 — Sentinel Prime AI sensitivity (Week 4). Penetration rate dominates the range; ASP second. Same ordering held for all five verticals.

Source in repo: `notebooks/week04_market_sizing.ipynb` · `reports/week04/market_sizing_workbook.xlsx` (tab per vertical) · `reports/week04/assumptions_register.csv` · tornado charts in `reports/week04/`

4. Competitive Landscape

Week 1 mapped 40+ companies across the five verticals; Week 2 narrowed to 15 priority peers (three per vertical) and profiled each on real, verifiable evidence: granted patent numbers from Justia and Google Patents, employee counts from filings and Revelio/CB Insights snapshots, and funding history from press announcements.

The finding that reframed everything else

Capital raised and defensible IP diverge sharply. The most heavily funded companies in the humanoid segment are not the most patent-dense, and several quieter competitors hold stronger, earlier, more specific grants. This matters because funding is the metric the trade press reports and the one that shapes internal anxiety — and it is the wrong one. A competitor with \$1bn and a thin patent position is buying time; a competitor with \$100M and a dense position around a specific mechanism is buying ground.

This was a hypothesis from Week 2's evidence. Week 6 tested it from the financial side and it held (Section 6): the durable businesses are the ones with defensible margin structures, not the ones with the biggest raises.

The 15 priority peers

Vertical	Competitor	Positioned against	Why it made the shortlist
Eldercare	Intuition Robotics (ElliQ)	Fari	Closest direct peer to Fari on eldercare companionship and social engagement; same tabletop companion form factor.
Eldercare	Labrador Systems	Fari	Eldercare peer attacking the same demographic from a physical-assistance angle, complementary to Fari's engagement foc...
Eldercare	OhmniLabs	Fari	Relevant to Fari's connection/engagement dimension via remote-presence, a different technical path to the same lonelin...
Education	Miko (Emotix)	Senpai	Direct peer for Senpai on AI learning companion and child-facing robotics; strongest consumer-education overlap.
Education	UBTECH Robotics	Senpai	Comparator for Senpai where education robotics meets a deep humanoid/AI portfolio and very strong IP.
Education	LEGO Education	Senpai	Incumbent benchmark for Senpai on classroom procurement, curriculum integration and brand trust.
Indoor Security	Knightscope	Sentinel Prime AI	Direct peer for Sentinel Prime AI and Aido Rover; most visible public-market security-robotics comparable.
Indoor Security	Cobalt Robotics	Sentinel Prime AI	Closest indoor-security analogue to Sentinel Prime AI; the human-in-the-loop model is a useful contrast point.
Indoor Security	RAD (AITX)	Sentinel Prime AI	Security-hardware breadth comparable to InGen's security ambitions; public micro-cap with disclosure to mine.
Outdoor Patrol	Asylon Robotics	Aido Rover	Closest peer to Aido Rover on outdoor perimeter security; multi-modal (ground+air) is a strategic contrast.
Outdoor Patrol	Boston Dynamics (Spot)	Aido Rover	Best-known legged platform; benchmark for Aido Rover on outdoor mobility and the deepest patent moat in the set.
Outdoor Patrol	ANYbotics	Aido Rover	Industrial-inspection adjacency to Aido Rover; shows the inspection-vs-security positioning split in outdoor robotics.
Humanoid	Figure AI	Aido Humanoid	Direct peer for Aido Humanoid and the clearest case of capital >> IP — a key strategic lesson for InGen.
Humanoid	Agility Robotics	Aido Humanoid	Direct peer for Aido Humanoid with the clearest logistics use case and genuine locomotion IP.
Humanoid	Unitree Robotics	Aido Humanoid	Comparator for Aido Humanoid on cost positioning; also overlaps Aido Rover via quadrupeds.

Three per vertical, each anchored to the InGen product it actually competes with, each with a dated retrieval and a written selection rationale. The dossier carries taglines, claimed capabilities, target customers, partnerships and the most recent public signal per company.

Evidence discipline

A note on what is *not* in the competitor dossier: real-time hiring counts. I could not verify them to a single dated snapshot, so rather than publish a number that looked precise and was not, R&D scale is represented by real headcount plus real patent counts, both dated. Several competitor claims about deployment scale were likewise excluded because the only source was the company's own marketing.

Source in repo: notebooks/week01_market_mapping.ipynb · notebooks/week02_competitive_intel.ipynb · data/week02/peer_positioning.csv · data/week02/patent_activity.csv · reports/week02/competitive_intelligence_dossier.pdf

5. Demand Signals

Market sizing says how big a prize is. It says nothing about whether interest is moving now. Week 5 built a demand-signal index from three independent public families — search interest (Google Trends), news volume and tone (GDELT), and voice-of-customer themes mined from public reviews (TF-IDF + KMeans) — combined with documented weights: search momentum 40%, news 35%, sentiment 25%.

Rank	Product	Vertical	Demand index
1	Aido Rover	outdoor_patrol	64.0
2	Fari	eldercare	59.2
3	Senpai	education	57.8
4	Aido Humanoid	humanoid	46.4
5	Sentinel Prime AI	indoor_security	40.7

The inversion

The demand ranking is close to the inverse of the market-attractiveness ranking. Aido Rover leads on attention (64) while Sentinel Prime AI — the vertical with the clearest near-term economics — comes last (41). These are not contradictory measurements; they are different questions. Search and news measure *attention*. Sizing measures *money*. The gap between them is where a commercial opportunity usually hides: **Sentinel looks like a strong asset with weak share of voice**, which is a marketing problem — a solvable one — rather than a product problem.

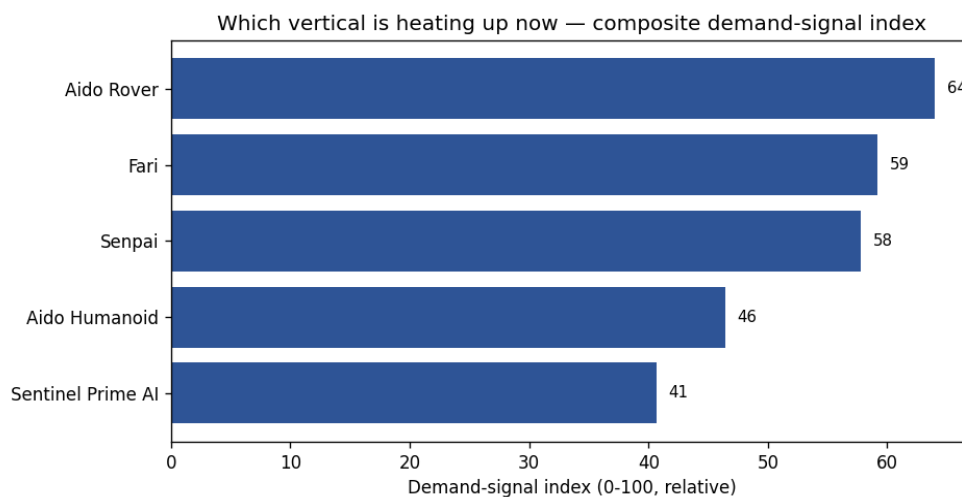


Figure 5.1 — Demand-signal index by product (Week 5). Relative, not absolute: the index compares verticals to each other, not to an external benchmark.

Pain points

Review mining produced five recurring themes per vertical with at least three verbatim examples each. Two themes recur across every vertical: reliability of autonomous navigation in cluttered real-world spaces, and the gap between demo behaviour and sustained field behaviour. Both are engineering-credibility issues rather than feature gaps, and both point at the same commercial answer — publishable field-reliability data.

Source in repo: `notebooks/week05_demand_signals.ipynb` · `data/week05/demand_signal_index.csv` · `data/week05/pain_points_long.csv` · `reports/week05/demand_signal_methodology.pdf` (weights and their justification)

6. Financial Benchmarks

Week 6 benchmarked InGen's competitive set against real, filed financials. Public peers come from FY2024 SEC filings and earnings releases; private peers from press-announced funding rounds, each with a source per row. Estimated cells are flagged as estimates, and EV/Revenue is deliberately left to compute from a live quote rather than frozen into the workbook.

Company	FY2024 revenue	Gross margin	Operating margin	Read
iRobot	\$682M	20.9%	-15.1%	Consumer robotics pure-play — losing money at scale
Teradyne	\$2,820M	58.5%	+20.5%	Test & measurement — the margin profile robotics aspires to
Cognex	\$915M	68.0%	+7.0%	Machine vision — highest margin; a components business
Symbotic	\$1,790M	18.0%	-5.0%	Warehouse automation — revenue scale, thin margin, systems-integration economics

Private capital

Company	Disclosed capital raised
Figure AI	\$1,675M
Apptronik	\$403M
Agility Robotics	\$400M
Neura Robotics	\$123M
1X Technologies	\$100M
Gecko Robotics	not disclosed in sourced rounds

What the financials say

The profitable peers are not robotics companies in the way the press means it. Teradyne (58.5% gross margin, +20.5% operating) and Cognex (68% gross margin) sell precision instruments and vision components into industrial customers. The company closest to InGen's consumer-adjacent shape — iRobot — ran a 20.9% gross margin and a -15.1% operating margin in FY2024. Symbotic has revenue scale (\$1.79bn) on an 18% gross margin, the signature of systems integration rather than product.

This is the Week 2 finding, restated in dollars. Figure AI's ~\$1.68bn raise is roughly 2.5× iRobot's entire FY2024 revenue — capital is not the constraint in this industry, and it is not the moat either. What separates the durable from the fragile is **margin structure**: whether you own something specific enough to charge for. For InGen the strategic read is that Sentinel-style deployments, sold against a costed guard alternative, have a defensible price story; volume consumer hardware, on this evidence, does not.

A caution on private valuations: the figures above are negotiated post-money numbers, not market prices. They tell you what one investor agreed to at one moment under one set of preferences and liquidation terms. They are not comparable to public market caps and the workbook does not treat them as such.

Source in repo: notebooks/week06_financial_benchmark.ipynb · reports/week06/peer_financial_workbook.xlsx (public / private / comps / summary tabs) · src/financials/peer_data.py (source per row, RETRIEVED dates) · reports/week06/benchmark_one_pager.pdf

7. The BI Platform

Phase 3 turned the analysis into infrastructure: a star-schema warehouse an analytics team could actually query, and dashboards on top of it across three BI tools.

The warehouse

Three fact tables — fleet telemetry (one row per robot per day), support tickets (one row per ticket), sales pipeline (one row per opportunity) — against four shared dimensions: date, product, customer, geography. Surrogate keys throughout, documented grain per fact, and foreign-key integrity verified on every load (zero orphans). It is populated with **100,000 synthetic telemetry rows**, **5,000 tickets** and **1,000 opportunities** generated from a fixed seed.

Why synthetic, and why it still matters: no internal data was available, so the warehouse cannot tell you anything about InGen's actual fleet. What it does is fix the *shape* — the schema mirrors what an InGen analytics team would genuinely hold. That makes the 15-query reference library and every dashboard built on it portable: point them at real data with the same shape and they work unchanged. The SQL self-assessment (12/12 across joins, window functions, CTEs, set operations and aggregations) was run against this warehouse.

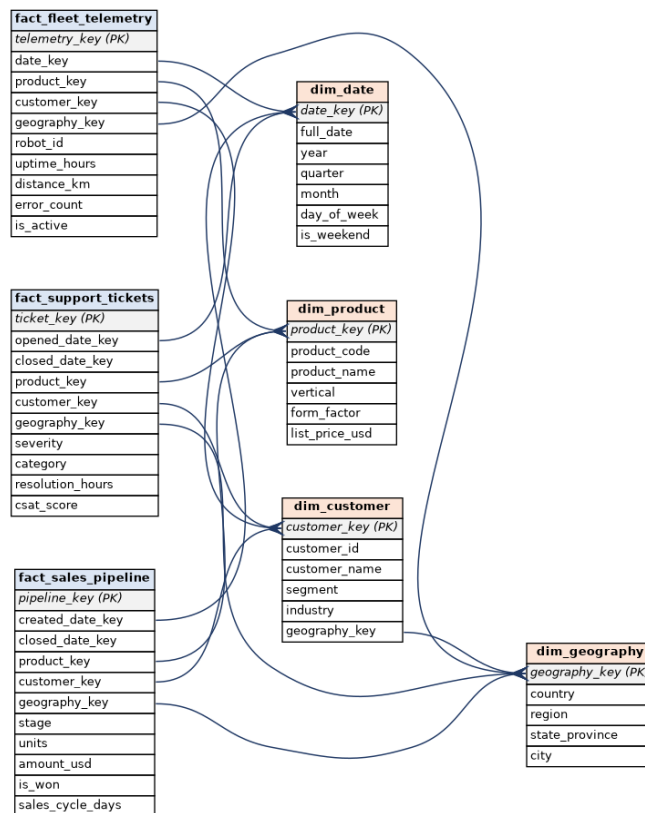


Figure 7.1 — Star schema (Week 7). Three facts, four dimensions, surrogate keys throughout.

Source in repo: notebooks/week07_sql_self_assessment.ipynb · src/warehouse/schema.sql · src/queries/reference_queries.sql (15 queries) · reports/week07/sql_score_sheet.md

Dashboards

Three tools, three different jobs. **Tableau Public — Market & Competitive:** the external-facing view built from Weeks 4–6 public data (market sizing, demand index, peer financials), five tabs, shareable outside the company. **Looker Studio — Product Analytics:** the operational view on the Week 7 warehouse — fleet health, support performance, sales pipeline, with date-range and product filters. **Power BI — Executive Scorecard:** eight traffic-light KPIs against targets with drill-through, backed by ten documented, reusable DAX measures.

Design was reviewed, not assumed: the palette is Okabe-Ito (colour-blind safe, and an earlier red/green draft was rejected for exactly that reason), chart types were checked against what they encode (a donut was replaced with ranked bars), and dashboards are fed pre-aggregated extracts — largest ~3.6k rows — so they load in well under five seconds rather than dragging 100k rows into the browser.



Figure 7.2 — Product Analytics dashboard (Week 8), rendered from the real Week 7 warehouse aggregates. Prototype: publishing to the live Looker/Tableau/Power BI services is done from the analyst's own accounts and was not completed within the internship.

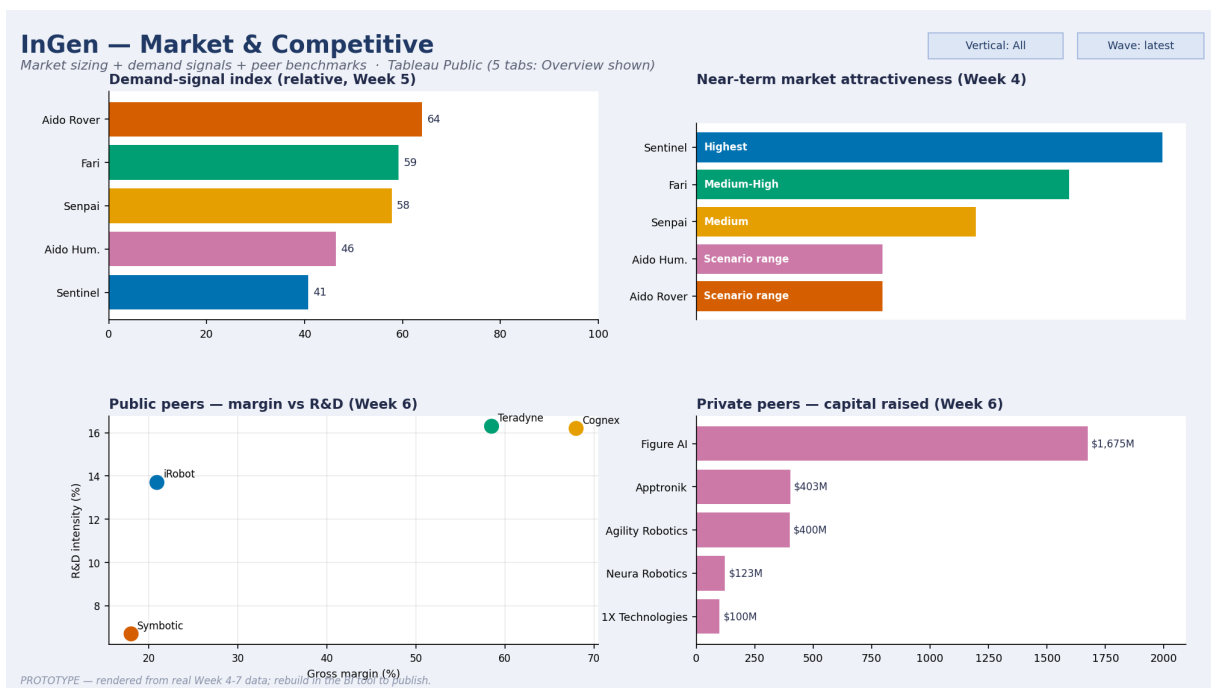


Figure 7.2a — Market & Competitive dashboard (Week 8), built from Weeks 4-6 public data.

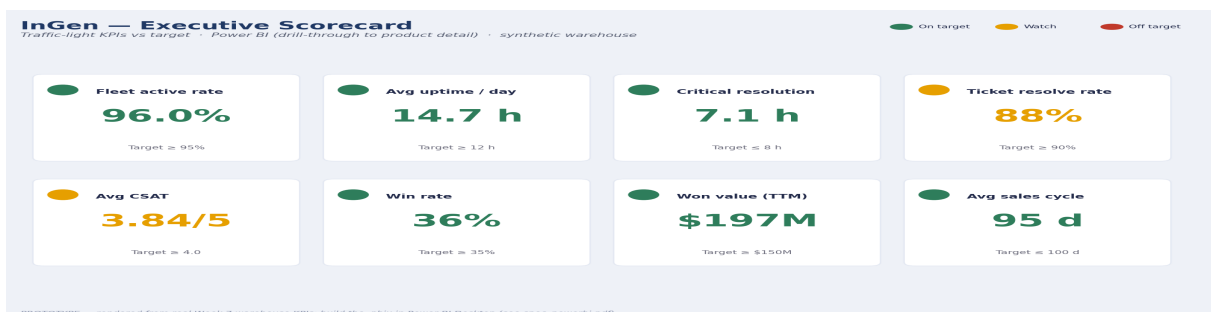


Figure 7.3 — Executive scorecard (Week 9). Two KPIs sit amber on the synthetic data — ticket resolve rate 88% against a 90% target, and CSAT 3.84 against 4.0.

Source in repo: reports/week08/spec_market_competitive.pdf · reports/week08/spec_product_analytics.pdf · reports/week08/design_review_log.md · reports/week09/spec_powerbi.pdf · docs/dax_measures.md (10 measures) · data/week08/extracts/

8. Forecasts

Week 10 turned Week 5's demand signals into 24-month forecasts. For each vertical, five models competed on identical data: three baselines (seasonal-naïve, ETS, ARIMA) and two stronger candidates (Prophet, and XGBoost on engineered features — lags, rolling momentum, news cadence and tone, peer-funding intensity). All were backtested on a held-out final twelve months and scored on MAPE and MASE.

Vertical	Product	Best model	MAPE	MASE	24-month estimate
eldercare	Fari	ETS	4.1%	1.00	49
education	Senpai	Prophet	3.0%	0.54	57
indoor_security	Sentinel Prime AI	XGBoost	5.7%	0.96	45
outdoor_patrol	Aido Rover	Prophet	2.5%	0.69	64
humanoid	Aido Humanoid	ETS	2.5%	0.59	67

MASE below 1.0 means the model beats a seasonal-naïve forecast — the honest bar. Four of five clear it; eldercare sits at 1.00, i.e. no better than naïve, and is reported as such rather than quietly dropped. **No single model dominates:** ETS wins twice, Prophet twice, XGBoost once. That is itself a finding — it argues for keeping a small model portfolio and re-selecting per series, not standardising on one algorithm.

What these forecasts are — and are not

The target is **search-interest momentum**, an index from 0 to 100. It is a *relative demand read*, not a unit forecast. Nothing here says how many robots InGen will sell; it says which verticals are gaining or losing attention relative to their own history, with quantified error bars. Any planning use requires calibration against internal sales data first.

Aido Humanoid (humanoid) — 24-month demand forecast

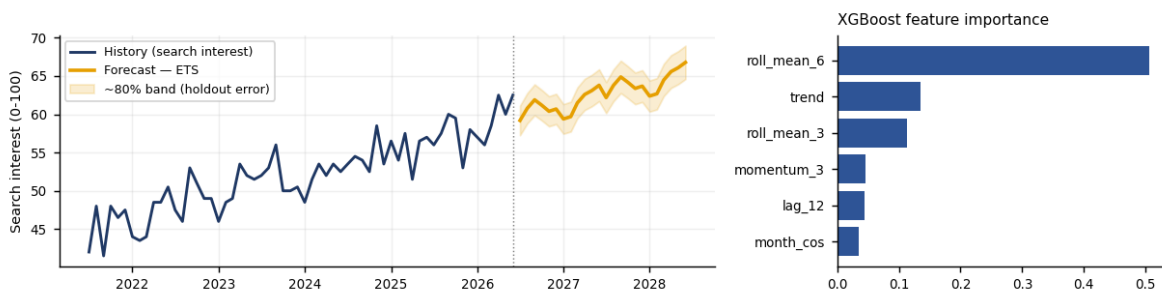


Figure 8.1 — Aido Humanoid: 24-month forecast with an ~80% band from holdout error, and the XGBoost feature importances for the same series (Week 10).

Humanoid adoption — Bass diffusion

For humanoid, momentum is not the interesting question — adoption timing is. I fitted a Bass diffusion curve to publicly stated shipment guidance from Goldman Sachs Research (January 2024): roughly 20,000 humanoid units shipped in 2025, a base case above 250,000 in 2030, and about 1.4 million units annually by 2035. Fitting in log space ($p=0.0018$ $q=0.520$ $M=11273308$) reproduces all three anchors exactly with a smooth S-curve, implying a cumulative installed base of about 4.3M by 2035.

As a cross-check, Morgan Stanley has projected roughly 13 million humanoids in service by 2035 — an order of magnitude above the Goldman-anchored cumulative figure. **That spread is the finding.** Two credible institutions differ by ~3× on the same decade, which is exactly why Section 3 treats humanoid as a scenario rather than a forecast. The curve is useful for shape — when the knee arrives — not for level.

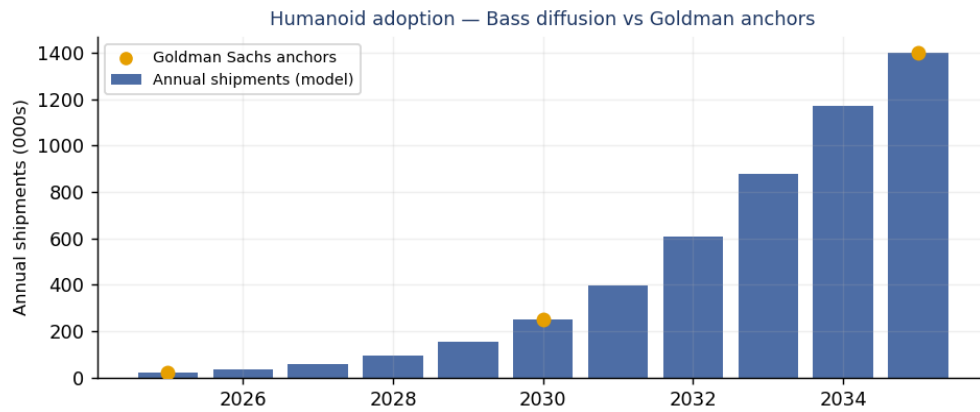


Figure 8.2 — Bass diffusion fit to Goldman Sachs shipment anchors (Week 10). Orange points are the published anchors; bars are the fitted annual shipments.

Source in repo: notebooks/week10_forecasting.ipynb · data/week10/forecast_results.csv · data/week10/bass_humanoid.csv · src/forecast/bass.py (anchors and citations in the module docstring) · reports/week10/forecast_report.pdf (one page per vertical)

9. Anomaly Detection Prototype (Sentinel Prime AI)

Sentinel Prime AI and Aido Rover both depend on anomaly detection. Week 11 prototyped that workflow on **two real, licensed public benchmarks** — no InGen data — and, more importantly, pushed the results through to the decision a product team actually has to make: what threshold do we ship?

Benchmark	What it is	Licence
NAB	realKnownCause/machine_temperature_system_failure.csv	MIT License (Copyright 2014-2024 Numenta Inc.)
SKAB	valve1/0.csv, valve1/1.csv, valve1/2.csv	GNU General Public License v3 (GPL-3.0)

Five detectors, identical splits

Dataset	Model	Precision	Recall	F1	Avg precision
NAB	AutoEncoder	0.425	0.800	0.555	0.512
NAB	Isolation Forest	0.349	0.800	0.486	0.499
NAB	Control chart (EWMA)	0.253	0.800	0.385	0.323
NAB	One-Class SVM	0.237	0.800	0.365	0.475
NAB	LOF	0.174	0.800	0.286	0.261
SKAB	AutoEncoder	0.403	0.846	0.546	0.436
SKAB	LOF	0.412	0.805	0.545	0.531
SKAB	One-Class SVM	0.339	1.000	0.506	0.339
SKAB	Control chart (EWMA)	0.360	0.800	0.496	0.405
SKAB	Isolation Forest	0.346	0.842	0.490	0.346

Isolation Forest, One-Class SVM, LOF and a small AutoEncoder (all PyOD) were compared against an EWMA control-chart baseline — the thing a plant engineer would already do, and the bar that justifies any added complexity. **AutoEncoder leads on NAB (F1 0.555), LOF on SKAB (F1 0.545); both clear the baseline.** Detectors were fitted unsupervised on a clean warm-up window — mirroring how a Sentinel unit would be baselined at commissioning — and labels were used only to evaluate.

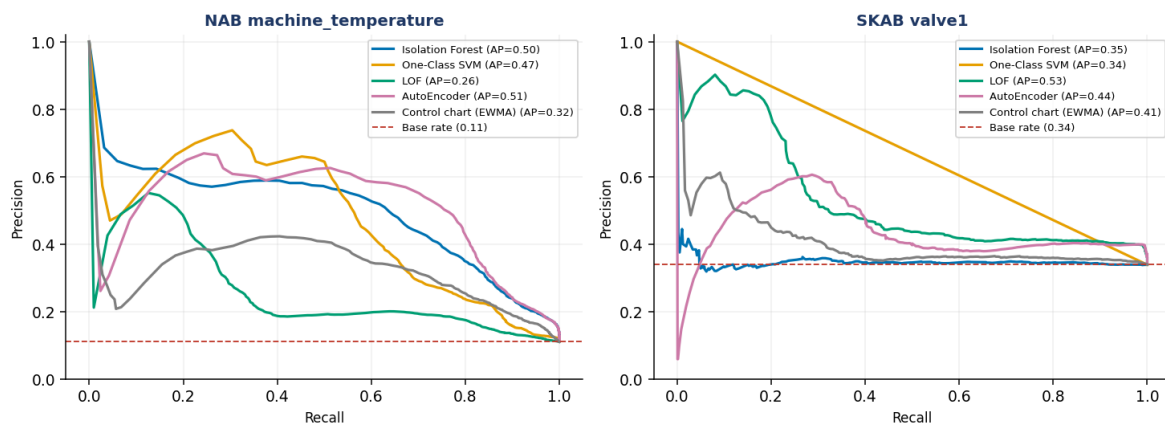


Figure 9.1 — Precision-recall curves, both benchmarks (Week 11). The dashed line is the base rate a random detector would achieve.

The finding: you cannot threshold your way out of a model-quality problem

The stated bars were a **recall floor of 80%** and a **false-alarm budget of 5%**. At 80% recall the point-level false-alarm rate bottoms out near **13%** — and it stays there across every threshold and persistence setting tested. That ceiling is set by the model's discriminative power (average precision ≈ 0.51), not by the threshold policy. **The 5% budget is unreachable at that recall, and the report says so rather than quietly relaxing the bar.**

But the point-level rate is the wrong metric. An operator does not experience '13% of readings'; they experience how often the phone buzzes. Counting one alert per contiguous run — the way a product actually pages — a **120-minute persistence filter cuts alert load 33×, from 16.35 to 0.50 false alerts per day**, while recall *ris*es to 88% and all 4/4 real failures are still caught. Real failures are sustained; false positives are isolated spikes.

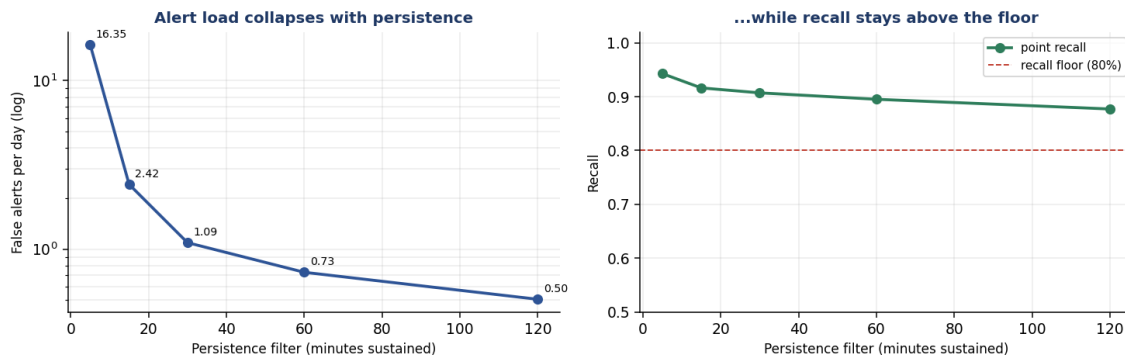


Figure 9.2 — Alert load collapses with persistence while recall holds above the floor (Week 11).

Adaptive thresholds lost, and the reason is the lesson

A rolling-quantile threshold cut false alarms hard (13.4% → 3.3%) but collapsed recall (80% → 33%). The diagnosis: NAB's failure windows run about two days, so a baseline computed over recent history **absorbs the anomaly into itself** and raises the threshold exactly when it should hold. This was not a tuning artefact — it survived a sweep of windows and EWMA rates. Freezing the baseline during an alert confirmed the diagnosis (recall recovered to 83%) but over-corrected, sending false alarms to 40%. **Verdict: adaptive thresholding pays only when drift is slower than the anomalies; for sustained failures, fixed threshold plus persistence is simpler and better.**

Recommendation for the product: two tiers. A fast tier (high cut-off, no persistence) for unambiguous events like intrusion, accepting more false alarms by design; and a sustained tier (120-minute persistence) for degradation, running at roughly half a false alert per day. One threshold cannot serve both, because persistence buys quiet at the cost of latency.

Transfer caveat: these thresholds are fitted to a machine-temperature signal. They transfer to Sentinel as a *method*, not as constants — they must be re-derived on real Sentinel telemetry before any field use.

Source in repo: `notebooks/week11/week11_anomaly_detection.ipynb · data/week11/results.csv · data/week11/operational_frontier.csv · data/week11/dataset_licenses.csv · reports/week11/sentinel_operational_framing.pdf`

10. Process Optimisation (Fleet Support)

Week 12 asked the operational counterpart to Phase 2's strategic questions: how do we run the support business better? The Week 7 warehouse carries ticket open/close dates and a total resolution time, but no stages, teams or workload — the three things the question needs. So the lifecycle layer was generated by a documented discrete-event model that **replays Week 7's real ticket stream**, with reconciliation enforced and tested: every ticket joins 1:1 back to the warehouse, and stage durations sum exactly to each ticket's cycle time.

Where the time goes

Stage	% of process time	Cumulative	Of which waiting
Parts & dispatch	53.2%	53.2%	100%
Remote diagnosis	22.3%	75.5%	31%
On-site repair	14.8%	90.3%	48%
Verification	5.7%	96.1%	27%
Triage	3.9%	100.0%	47%

Parts & dispatch alone is 53% of all process time, and 100% of it is unstaffed waiting — nobody is working during it, so no amount of headcount shortens it. On-site repair, by contrast, is 48% queueing, which headcount *can* shorten. That distinction drives the entire recommendation.

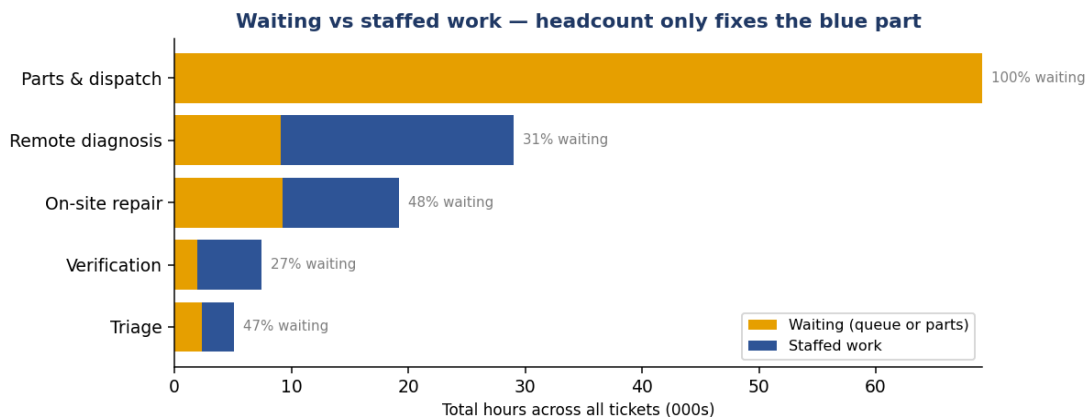


Figure 10.1 — Waiting versus staffed work by stage (Week 12). Headcount only shortens the blue.

What actually drives cycle time

A regression of log cycle time on the drivers you would instinctively reach for — product, severity, geography, weekday, team workload — explains just **4%** of the variance ($R^2=0.042$, $n=5,000$). Severity and queue length are real and significant (a Critical ticket clears ~40% faster; each ticket already queued adds ~10%), but second-order. Add one structural fact — did this ticket need a physical part? — and **R^2 jumps to 0.72**: a dispatched ticket takes **+578% longer** [+555%, +601%]. **Support cycle time is not a people-management problem; it is a parts problem.**

What each fix would buy

Change	Effect on mean cycle time	95% CI	Verdict
+1 Field Ops FTE	-7.5%	[-2.41h, -1.50h]	significant
Reroute Tier-1 triage	-1.9%	[-1.02h, 0.05h]	no measurable effect
Regional parts buffer (EU+APAC)	-22.9%	[-6.52h, -5.39h]	significant
Both together	-29.1%	[-8.02h, -7.09h]	significant

Eight replications per scenario, paired to identical seeds. **The regional parts buffer is worth roughly three times the field engineer** (–23% vs –8%), and it pulls the P90 from 64 hours to 47 — the tail is what customers actually

complain about. Rerouting Tier-1 triage — the obvious lever, and the one the brief suggested — **does nothing measurable**, because triage is 4% of process time. Adding the field engineer nearly eliminates field-queue wait (4.9h → 0.4h) yet buys only 8% overall: a textbook Amdahl's-law ceiling, since that stage is ~15% of the process.

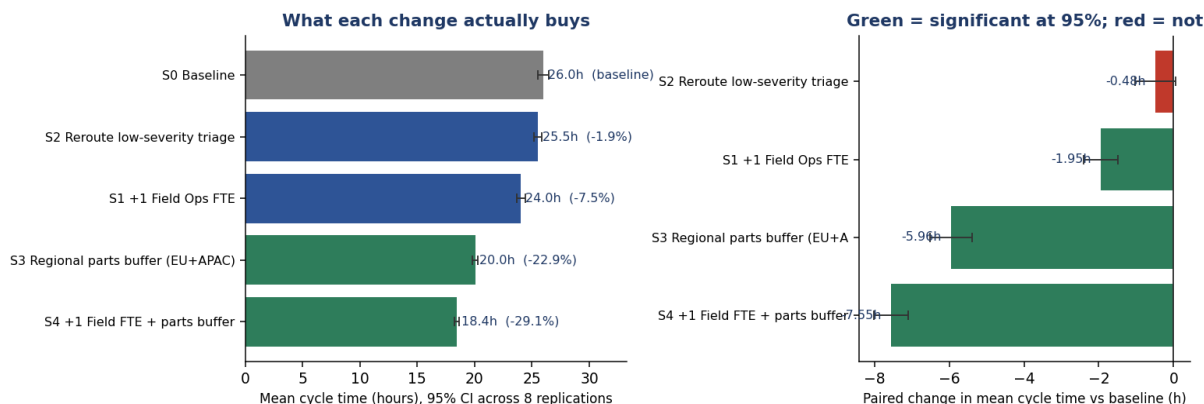


Figure 10.2 — Simulated operational changes with 95% CIs (Week 12). Green = significant.

Trade-offs, named: the parts buffer ties up working capital in inventory at two sites and carries obsolescence risk as hardware revs — but it is the biggest tail-improvement, so the best CSAT lever. The field engineer is a recurring salary against a benefit capped by the stage's share of the process. Rerouting triage costs engineering time for no measurable return.

Source in repo: `notebooks/week12/week12_process_optimization.ipynb` · `src/process/process_model.py` (documented parameters) · `data/week12/stage_decomposition.csv` · `data/week12/capacity_sim_paired_deltas.csv` · `reports/week12/process_optimization_memo.pdf`

11. Limitations

This section is the honest inventory of what public data could not answer. It is deliberately specific: each item names what is missing, what it affects, and what would fix it.

1. No internal InGen data was used — anywhere

Every figure in this report comes from public sources or from clearly-labelled synthetic data. That was the brief, and it has a cost. Market bottom-ups rest on assumed penetration rates rather than InGen's observed conversion; demand signals are a proxy for interest, not measured pipeline; the entire product-analytics layer (Sections 7 and 10) is a synthetic warehouse. **These analyses are directional, not decision-grade**, and should not be used to set targets without calibration.

2. The synthetic warehouse fixes shape, not fact

Weeks 7–9 and 12 rest on 100k synthetic telemetry rows, 5k tickets and 1k opportunities. The schema is realistic; the numbers are generated. Nothing in Sections 7 or 10 is a measurement of InGen's fleet, support org or pipeline. The KPI targets on the executive scorecard are my reasonable assumptions, not InGen's actual targets — I asked for the real ones and did not receive them within the internship.

3. Live data sources were not reachable from the build environment

The Google Trends, GDELT and SEC EDGAR collectors are real code written against real APIs, but the sandbox could not reach them, so several weeks ran with clearly-labelled synthetic fallbacks and deterministic seeds. **Week 11 is the exception and shows what changes when real data is available**: NAB and SKAB were downloaded from source, and the results in Section 9 are measurements on real sensor data. The fix is a networked run — the pipelines are already written.

4. Forecasts are momentum reads, not unit forecasts

Section 8's target is a 0–100 search-interest index. It answers 'is attention rising here, relative to its own history?' It does not answer 'how many units will we sell?'. The Bass curve inherits the uncertainty of its anchors — and Goldman and Morgan Stanley differ by roughly 3× on humanoid by 2035, which is why humanoid is scenario-bounded throughout.

5. Published market estimates disagree by an order of magnitude

This is a property of the industry, not of the analysis. Humanoid ~\$290M to ~\$4.89bn; security robotics ~\$4.7bn to ~\$19bn. Every market figure in Section 3 is a sourced range for that reason. A reader who wants a single number is asking the data for something it does not contain.

6. The three dashboards are built but not published

Specs, upload-ready extracts, prototypes and the design-review log are complete for Tableau, Looker Studio and Power BI. Publishing to the live services runs through the analyst's personal accounts and was not completed. There are no live URLs, and none are claimed.

7. Week 12's process findings are properties of a model

The support lifecycle was generated by a documented discrete-event simulation, because the warehouse had no stage data. Team sizes were chosen to load the teams realistically against the observed arrival rate, not observed from data. The driver regression was validated against the simulation's own known parameters (7 of 9 effect directions recovered), so the *method* is sound — but the –23% parts-buffer number is a simulation result, not a measurement. No shift calendar is modelled, so the null weekday result is not evidence that weekday does not matter in reality.

8. Things I could not verify, and therefore left out

Real-time competitor hiring counts (no dated snapshot I could stand behind), several competitor deployment-scale claims (only source was company marketing), and precise EV/Revenue multiples (deliberately left to compute from a live quote rather than frozen into a workbook that would silently go stale).

12. Recommended Next Steps

For the business

Priority	Recommendation	Why, from this work
1	Market Sentinel Prime AI harder	It ranks first on near-term market attractiveness and <i>last</i> on demand signal — the widest gap in the portfolio. Strong asset, weak share of voice: a marketing problem, which is a solvable one. (§3, §5)
2	Fund a regional parts buffer before hiring field engineers	–23% mean cycle time versus –8% for +1 FTE, and the P90 falls from 64h to 47h. Parts is 53% of process time and 100% waiting; headcount cannot touch it. (§10)
3	Ship Sentinel anomaly detection with two-tier alerting	A fast tier for intrusion, a 120-minute sustained tier for degradation at ~0.5 false alerts/day. One threshold cannot serve both. (§9)
4	Treat humanoid as an option, not a plan	Credible analysts differ ~3× on 2035. Scenario-bound the investment; revisit when the Bass curve's knee is observable rather than projected. (§3, §8)
5	Publish field-reliability data	The two pain-point themes recurring across every vertical are navigation reliability and the demo-versus-field gap. Both are credibility problems that published reliability data addresses directly. (§5)

For the analytics function

The single highest-leverage action is granting read access to one slice of real internal data — even a small anonymised export of fleet telemetry or sales-pipeline records. The synthetic warehouse already matches the intended schema, so real data slots in with minimal rework, and it dissolves most of Section 11 at once: the warehouse becomes real, the dashboards become operational, Week 12's diagnostic becomes a measurement rather than a simulation, and the forecasts gain a calibration target.

Beyond that, in order: (a) run the Week 3/5/6 collectors in a networked environment — the code is written and tested, it just needs egress, and Week 11 demonstrates the quality jump when real data is available; (b) publish the three dashboards and circulate the URLs; (c) confirm the real KPI targets so the executive scorecard's traffic lights mean something; (d) re-derive Week 11's thresholds on real Sentinel telemetry before any field use.

What a successor inherits

Thirteen self-contained weekly modules, each with a README, a plan, a status note and — where applicable — a passing test suite. A reusable public-data ingestion pipeline across twelve government and institutional sources. A star-schema warehouse with a 15-query reference library. A forecasting harness that scores any new model against naive baselines automatically. An anomaly-detection bench that runs five detectors on licensed public data. A discrete-event simulator of the support process. **HANDOFF.md** in the repository root describes how to pick up each workstream, with known issues and TODOs.

Source in repo: [HANDOFF.md](#) · [INDEX.md](#) · [README.md](#) · [reports/week13/](#)

Appendix A — Repository Map & Reproduction

Every number in this report is reproducible from a clean checkout. Each week is a self-contained module with its own README, plan, status note and (where applicable) test suite; INDEX.md links them all.

Where each section's evidence lives

Section	Notebook / workbook	Key outputs
\$3 Market Sizing	notebooks/week04_market_sizing.ipynb	reports/week04/market_sizing_workbook.xlsx; assumptions_register.csv; tornado_*.png
\$4 Competitive	notebooks/week01_market_mapping.ipynb; week02/week02_pipeline_monitoring.ipynb	data/week02/competitive_intelligence_monitoring.csv; patent_activity.csv; competitive_intelligence_dossier.pdf
\$5 Demand Signals	notebooks/week05_demand_signals.ipynb	data/week05/demand_signal_index.csv; pain_points_long.csv; demand_signal_methodology.pdf
\$6 Financials	notebooks/week06_financial_benchmark.ipynb	reports/week06/peer_financial_workbook.xlsx; src/financials/peer_data.py (source per row)
\$7 BI Platform	notebooks/week07_sql_self_assessment.ipynb	src/warehouse/schema.sql; src/queries/reference_queries.sql; week08/09 specs; docs/dax/
\$8 Forecasts	notebooks/week10_forecasting.ipynb	data/week10/forecast_results.csv; bass_humanoid.csv; reports/week10/forecast_report.pdf
\$9 Anomaly Detection	notebooks/week11/week11_anomaly_detection.ipynb	data/week11/results.csv; operational_frontier.csv; sentinel_operational_framing.pdf
\$10 Process Optimisation	notebooks/week12/week12_process_optimization.ipynb	data/week12/stage_decomposition.csv; capacity_sim_paired_deltas.csv; process_optimization.pdf

Reproducing the analysis

```
pip install -r requirements.txt
```

```
# Week 3 – public-data pipeline and warehouse
```

```
python -m src.ingest.run_all # offline fallbacks by default
```

```
INGEST_ALLOW_NETWORK=1 python -m src.ingest.run_all # live public APIs
```

```
python -m src.ingest.build_warehouse
```

```
python -m pytest src/ingest/tests -q # 15 tests
```

```
# Week 7 – analytics warehouse
```

```
python -m src.generator.generate_synthetic && python -m src.warehouse.build_warehouse
```

```
python -m src.queries.run_queries # 15 reference queries
```

```
# Week 10 – forecasting
```

```
python -m src.forecast.run_forecasts && python -m src.forecast.build_report
```

```
# Week 11 – anomaly detection (downloads NAB + SKAB, then caches)
```

```
python -m src.anomaly.run_all && python -m src.anomaly.build_report
```

```
python -m pytest src/anomaly/tests -q # 12 tests
```

```
# Week 12 – process optimisation
```

```
export WEEK7_DB=/path/to/week07/ingen_warehouse.duckdb
```

```
python -m src.process.stage_generator && python -m src.process.capacity_sim
```

```
python -m pytest src/process/tests -q # 12 tests
```

Determinism: Week 7 generator, Week 11 detectors (random_state=42) and Week 12 simulation (SEED=20260710) are all seeded. Re-running yields the figures in this report.

Appendix B — Data Sources & Licences

All public. Where a source required a key (US Census, FBI Crime Data Explorer), the collector reads it from an environment variable and falls back to a labelled offline fixture — no keys are committed.

Government and institutional statistics (Week 3 pipeline, 12 datasets)

Source	Used for
US Census Bureau — population by age	Eldercare addressable population (Fari bottom-up)
BLS — home health aides; security guards; warehouse employment	Logistics employment baselines for Fari, Sentinel, Aido Rover
NCES — K-12 enrolment	Education addressable base (Senpai)
OECD — long-term care spend	Eldercare spending envelope
World Bank — population 65+; education spend; R&D spend	Spice country sizing and R&D-intensity context
FBI Crime Data Explorer — property crime	Security demand baseline (Sentinel, Aido Rover)
OpenAlex — robotics publications	Research-intensity signal
SEC EDGAR — robotics filings	Public-peer financials (Week 6)

Market, demand and competitive sources

Source	Used for
Google Trends	Search-interest momentum — the Week 10 forecast target
GDELT	News volume and tone
Public product reviews	Voice-of-customer pain-point mining (TF-IDF + KMeans)
Justia / Google Patents	Granted patent counts per peer (Week 2)
Company filings, press releases, Crunchbase / CB Insights	Funding rounds, headcount, deployment claims
Goldman Sachs Research (Jan 2024)	Humanoid shipment anchors: ~20k (2025) → 250k+ (2030) → ~1.4M/yr (2035); \$38bn TAM by 2035
Morgan Stanley Research (Apr 2025)	Humanoid cross-check: ~13M units in service by 2035

Licensed public benchmarks (Week 11)

Benchmark	Series used	Licence	Citation
NAB	realKnownCause/machine_temperature_system_failure.csv	MIT License (Copyright 2014-2024)	Alvin Chan (2015), 'Evaluating Real-time Anomaly Detection Algorithms — the Numenta Anomaly Benchmark', IEEE ICMLA.
SKAB	valve1/0.csv, valve1/1.csv, valve1/2.csv	GNU General Public License v3 (GPLv3)	Rafael Kozitsin (2020), 'Skoltech Anomaly Benchmark (SKAB)', Kaggle.

Synthetic data — declared

Where	What	Why
Week 7 warehouse	100,000 telemetry rows; 5,000 tickets; 1,000 pipeline opportunities	Numbers (if not seeded) are deterministic; schema mirrors what an InGen analytics team would have
Weeks 8-9 dashboards	Product-analytics and scorecard figures drawn from the Week 7 warehouse	Week 7 warehouse targets are assumptions, not InGen's actual targets
Week 12 process layer	Stage events generated by a documented DES replaying Week 7's scenarios	Week 7's scenarios are stage area/workload fields, which the analysis requires
Weeks 3/5/6 fallbacks	Deterministic offline fixtures where live APIs were unreachable	Sandbox egress restrictions; collectors are real code against real APIs

Nothing synthetic is presented as real anywhere in this report or the repository.

Prepared by Ziheng Wang · Data Analyst Intern · inGen Dynamics (Futureonauts Program) · Week 13 capstone. All analysis rests on public data or clearly-labelled synthetic data; no InGen internal data was used. Every section cites the notebook or workbook that produced it, and the repository reproduces every number here from a clean checkout.